Quiz 5 – Answer Key

1. A, D, E
2. A, B, C, E
3. A, B, D
4. A, B, C, D
5. B, C, D
6. B
7. (a) ATE = 7.0
   (b) D = 13.0
   (c) ATE = 7.0**,** D = 13.0 → D ≠ ATE**.**
   Interpretation (1 mark): D overestimates the true average causal effect. Reason: treatment assignment is not random in this dataset — the treated individuals have higher expected outcomes absent treatment (their $Y(0)$or baseline potential is not lower enough to explain the whole gap). Thus the observed difference mixes the true causal effect with selection bias/confounding. To make Dequal to ATE we would need ignorability/random assignment (and SUTVA).